— THE GREEK POVERTY PARADOX

If Greece Has Recovered, Why Do So Many Households Still Struggle?

Official income poverty affects roughly one person in five. Yet about two households in three report difficulty making ends meet. The distance between them reveals a moving poverty line, an uneven recovery and the continuing weight of unemployment, wages and housing.

1 in 5

19.6%

People at risk of poverty — the official measure

2 in 3

66.7%

Households struggling to make ends meet

FIGURE 1 **Greece reports far more hardship than countries with similar income poverty**

DESCRIPTIVE Does official income poverty account for the hardship Greek households report?

▼ Methods and limits

Read with this. Descriptive comparison of two official measures. Reported hardship is answered by HOUSEHOLDS and income poverty counts PEOPLE, so the axis is labelled percent rather than either. In the second view the fitted line excludes Greece, describing the European pattern Greece is being judged against rather than one Greece helped set. The cross marks the median country on each measure taken SEPARATELY, so it is a reference point rather than an actual country: no member state necessarily sits there. Both views are country-level and say nothing about any individual household.

How Greece's hardship gap developed

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: reported hardship	77.7	76.8	77.2	74.1	71.0	69.8	71.0	68.4	67.0	66.7
EU median: reported hardship	28.9	25.9	22.6	20.5	18.5	16.8	15.4	17.9	16.6	17.1
Greece: income poverty	21.4	21.2	20.2	18.5	17.9	17.7	19.6	18.8	18.9	19.6
EU median: income poverty	16.3	16.5	15.9	16.4	15.4	16.1	15.7	15.6	15.4	15.5

Where countries stood in 2024

Series	Income poverty (%)	Reported hardship (%)
Czechia	9.5	14.2
Belgium	11.4	17.1
Denmark	11.6	12.8
Netherlands	12.1	7.3
Finland	12.6	8.9
Ireland	12.7	16.9
Slovenia	13.2	13.6
Poland	13.8	11.5
Austria	14.3	13.1
Hungary	14.3	19.8
Slovakia	14.5	28.7
Cyprus	14.6	20.8
Sweden	14.8	9.9
Germany	15.5	7.3
France	15.9	21.8
Portugal	16.6	21.8
Malta	16.9	17.3
Luxembourg	18.1	8.5
Italy	18.9	18.7
Romania	19.0	23.8
Greece	19.6	66.7
Spain	19.7	21.9
Estonia	20.2	9.9
Croatia	20.3	19.8
Lithuania	21.5	11.4
Latvia	21.6	23.3
Bulgaria	21.7	37.4

WHAT THE OFFICIAL NUMBERS MISS

Two official numbers that don't agree

Europe measures poverty in two ways, and both are official.

The first counts *people* whose household income falls below 60% of what a typical household in their country earns. It is a number about position: are you far behind your neighbours? The second is a question put directly to *households*: are you having difficulty making ends meet? It is a number about experience. The two count different things — people against households — which matters for reading them side by side, though it is not what makes them disagree this badly.

In most of Europe the two roughly agree. In Greece they are 52.6 percentage points apart, and have been for a decade.

Put plainly: roughly one Greek in five is officially counted as at risk of poverty. Roughly two households in three say they are struggling to get by. Those are not two attempts to measure the same thing that landed slightly apart. They are far enough apart that no ordinary measurement error, and no difference in who is being counted, closes them.

> Greek subjective hardship runs 52.6 points above relative income poverty on average, and Greece ranks first of 27 on hardship while ranking seventh on AROP.

Greece also ranks first in the European Union on the question about struggling, and seventh on the income measure. That combination is what makes this a puzzle rather than a ranking. And Greece is not simply the last country in a long queue: the distance between Greece and the next country is wider than the distance covering most of the rest of Europe. Whatever is happening here is not a stronger dose of what happens elsewhere.

This report is an attempt to find out what that gap is made of. The honest summary of the answer, given at the start so that nothing later reads as a reveal: we can account for part of it, we can rule some explanations out, and most of it remains unexplained.

A ruler that shrank

The official poverty line isn't fixed. It moves with the very economy it is supposed to be measuring.

Here is the quiet flaw at the centre of the EU's standard poverty measure: it is not an income level. It is a percentage — 60% of whatever the national median income happens to be, *that year*. In an ordinary economy, where incomes drift up slowly and roughly together, that is a reasonable way to define being poor relative to your neighbours.

Greece's economy, from 2010, was not ordinary. Incomes across the whole country fell together, hard and fast. And when the median falls, the poverty line falls with it. A household earning exactly what it earned five years earlier could find itself reclassified from poor to not poor.

> *"Not because anything in its life had improved, but because the ruler measuring it had shrunk to match the collapse around it."*

So the official income-poverty rate barely moved through the worst years. Not because Greek households weren't getting poorer — they were, dramatically — but because the yardstick was falling with them, always re-centring on a poorer and poorer normal.

What happens if you refuse to let the ruler move? Hold the line where it stood in 2008, adjust it only for inflation, and measured poverty roughly doubles: from under 20% before the crisis to a peak above 40% in 2014.

FIGURE 2 **Who counts as poor barely moved; what the poverty line buys fell by a fifth**

DESCRIPTIVE

What happened to the line Greek poverty is measured against, and what does a fixed line show instead?

▼ **Methods and limits**

Read with this. The anchored series is an approximation built in this project and is labelled as such wherever it appears. The first view counts PEOPLE, the second counts EUROS. In the second, both lines are the same official threshold: one as published in each year's own money, the other converted into what it could buy in 2008. The cash line recovers and the purchasing-power line does not, and that difference is why the first view's two counts diverge. Both views are Greece only and support no cross-country statement.

Who falls below a fixed line

Series	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Greece: fixed 2008 threshold	25.6	22.8	20.7	21.5	21.2	19.7	17.0	18.0	24.6	33.3	39.7	40.6	39.8	39.8	38.3	37.3	34.6	30.8	32.7	33.6	32.0	30.2	27.5
Greece: current-year threshold	20.7	19.9	19.6	20.5	20.3	20.1	19.7	20.1	21.4	23.1	23.1	22.1	21.4	21.2	20.2	18.5	17.9	17.7	19.6	18.8	18.9	19.6	19.6

What the line itself is worth

Series	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Threshold in cash terms	4923	5306	5650	5910	6120	6480	6897	7178	6591	5708	5023	4608	4512	4500	4560	4718	4917	5269	5251	5712	6030	6510	7020
Same threshold in 2008 purchasing power	5819	6089	6264	6343	6377	6480	6808	6768	6027	5168	4589	4270	4228	4216	4226	4338	4498	4884	4838	4815	4878	5113	5358

A word on where these numbers come from, since this chapter reaches back further than the rest. From 2010 onward the figures are Eurostat's own, published and used as they are. For the years before that Eurostat does not publish this measure, so the earlier values are built from its own components using a rule we checked against the official series everywhere the two overlap — 432 country-years, agreeing almost exactly. Those earlier years are used for the picture only. Nothing later in this report is tested on them.

> The outcome is Eurostat's official subjective-hardship indicator, extended backwards before 2010 using a constructed series validated against it on 432 overlapping country-years.
>
> *The limits of this.* pre-2010 provenance is ours, not Eurostat's.

Two cautions, because this chart is easy to over-read. This is a measure of Greece against its own past, and it contains no other country: it cannot tell you that Greece's line fell further than anyone else's. And the fixed line is not the *correct* line. The relative measure is doing exactly what it was designed to do — measuring position — and the fixed measure answers a different question about living standards. When a whole country falls together, those two questions come apart, and that separation is the point.

The wider net

Europe already knows income alone is too narrow. Its headline measure of social exclusion casts a wider net: it counts you if your income is low, or if you can't afford a list of ordinary things, or if the adults in your household are barely working. If the puzzle described at the start is just that the income measure is too narrow, the wider one should mostly dissolve it.

> Switching to AROPE closes 9.8 of those 52.6 points (19%), leaving 42.8 unexplained, and its contribution shrinks from 11.0 points in 2015 to 7.3 in 2024.

It helps, and it isn't enough. The wider net picks up under a quarter of the distance. More telling is the direction of travel: its contribution is getting *smaller*, from eleven points at the start of the period to seven by the end. As an explanation of this puzzle it is weakening, not strengthening.

There is also something hidden inside that headline number. It is built from three separate conditions joined by *or*: your income is low, or you can't afford a list of ordinary things, or the adults around you are barely working. Because they're joined by *or* and not added up, the headline can't be taken apart into neat slices — a household counted for two reasons is still one household.

What that hides is that the three don't move together. One of them can be falling while another rises, and the combined rate sits calmly in the middle looking like nothing much is happening. Nor do they apply equally to everyone: the work-intensity condition is defined over working-age adults, so a household of pensioners can't trigger it at all. Comparing the headline across age groups is therefore partly comparing which conditions were even eligible to fire.

This is a recurring shape in the whole investigation. The averages are calm. What's underneath them isn't.

The average hid a divide

National averages are averages of people, and people are not interchangeable.

Underneath a Greek headline rate that moved gently, the age groups did not move together at all. The pattern that emerges when you split them apart is the kind of thing an average is very good at concealing: some groups improving while others did not, and the combined figure sitting placidly between them.

This matters for the question this report is asking. A country where hardship has become concentrated in particular groups can look, on the headline, exactly like a country where it eased for everybody a little. The household answering the survey question is not answering about the national average. It is answering about itself.

It also complicates the recovery story that follows. Aggregate improvement in unemployment and consumption is real. Whether it reached the same people who absorbed the worst of the crisis is a different question, and the age split is the closest this data comes to an answer: not evenly.

We stop short of the stronger version of that claim. Showing that groups moved differently is not the same as showing which group's experience drives the national answer to the survey question, and this study cannot do the second.

A MATERIAL FOOTPRINT

Is any of it real?

This is the chapter where the whole thing could have collapsed.

If Greek households say they are struggling while nothing in their material circumstances corresponds to it, then this is a story about how people answer questions, not about poverty, and everything after this point measures a phantom. So it has to be dealt with before anything else.

The test is whether the reported difficulty moves together with things that name events rather than feelings: falling behind on bills, being unable to handle an unexpected expense, being unable to heat the home, going without several basic things at once.

> Reported difficulty co-moves with arrears, inability to meet an unexpected expense, inadequate heating and severe deprivation at within-country correlations of 0.63 to 0.80.
>
> *The limits of this.* same-instrument corroboration, NOT independent validation | all items and the outcome come from EU-SILC | not uniform: arrears within Greece is 0.371.

They move together, and they do so *within* countries, which is the harder test — it can't be satisfied by rich countries simply differing from poor ones.

Now the honest part, and it matters more than the result. Every one of those items comes from the same survey, asked of the same household, in the same sitting, as the question about making ends meet. A household in a grim mood about its finances will answer the whole set grimly, and that alone would produce numbers like these. This is one instrument agreeing with itself.

> ***"It is real evidence and it is not independent confirmation, and the difference between those two things runs through the rest of this report."***

It isn't even uniform. Falling behind on bills — the item you would expect to be the hardest, most factual anchor — tracks the reported difficulty far more weakly within Greece than the others do. Arrears require having credit and bills to fall behind on. A household that lost access to credit years ago, or never had it, can be in serious trouble without ever registering.

> Those same four items statistically absorb 71% of Greece's baseline residual (+46.92 to +13.74).
>
> *The limits of this.* ABSORPTION, never explanation | shared instrument is not shared cause | diagnostic only; never a headline explanation.

That word — absorb — is doing careful work, and it is not a synonym for explain. Put those deprivation items into the model and most of Greece's unexplained excess stops being statistically visible. That tells you the two things share a great deal of information. It does not tell you one causes the other, because both are measured by the same survey of the same households on the same day. The result that flips, later on, shows how much rides on that.

Good company for a bad number

The obvious first suspicion about a number that disagrees with every other number is that the number is broken.

So it is worth asking what company it keeps. Take sixteen separate measures of Greek life — wages, hours worked, prices, saving, what households expect of next year, how many people are leaving — the same sixteen every time, chosen only because both a 2008 and a 2024 reading exist for each, so growth in the count can't be explained by counting more things later. Then ask a blunt question of each one: is Greece in the worst fifth of Europe on this? Count how many say yes.

FIGURE 3 **On the same 16 indicators, Greece went from the EU's worst fifth on 4 in 2008 to 11 in 2024**

DESCRIPTIVE Did Greek disadvantage deepen on a few measures, or spread across many?

▼ Methods and limits

Read with this. Descriptive summary of the condition, not a predictor of it. DESCRIPTIVE ONLY. Breadth was tested as a predictor of reported hardship in P3a and does not survive: on its own it is not significant (p = 0.12), and once the other accumulated measures enter the model its coefficient reverses sign. It summarises the condition rather than explaining it. THE BASKET IS FIXED: these are the 16 indicators with a valid EU position in both 2008 and 2024, chosen without reference to whether they improved or deteriorated, so the same things are counted at both ends and in every year between. Unemployment, youth unemployment and the employment rate are absent because comparable EU coverage for them begins in 2009; hours worked, pay per hour, the work-effort squeeze and real wages are all present, so this is not a basket without labour-market information. The outcome and every model covariate are excluded. THE EU-COUNTRY MEDIAN is the median of the 27 member states' own shares on this basket -- a median across countries, not Eurostat's population-weighted EU aggregate -- named accordingly rather than simply 'EU'. Croatia never reaches full 16-indicator coverage in any basket year and does not appear as a line for that reason. The 'Which measures' view's axis is POSITION, not value: the indicators have no common unit, so 0 is the best place in the Union to be on that indicator and 100 the worst, whichever direction 'worse' runs in.

Of 16 indicators, **4 were already in the EU's worst fifth in 2008, 7 entered it by 2024, and none left**.

Which measures

Series	2008 position	2024 position	Transition
Hours worked against hourly pay	59.3	100.0	entered worst fifth
Prices measured against wages	48.1	100.0	entered worst fifth
Pay per hour worked	55.6	100.0	entered worst fifth
Household income after inflation	51.9	100.0	entered worst fifth
Wages after inflation	51.9	100.0	entered worst fifth
The poverty line after inflation	51.9	100.0	entered worst fifth
Keeping the home warm	73.1	98.1	entered worst fifth
Hours worked each week	94.4	100.0	already worst fifth
What households expect of the year ahead	100.0	100.0	already worst fifth
Citizens leaving the country	86.4	96.2	already worst fifth
Income inequality	80.8	81.5	already worst fifth
Food prices	14.8	77.8	outside both years
Economic output per person	55.6	77.8	outside both years
Prices overall	53.7	70.4	outside both years
Housing and energy prices	63.0	59.3	outside both years
What households actually spend	48.1	48.1	outside both years

How many measures

Series	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece	25.0	18.8	43.8	50.0	50.0	50.0	43.8	50.0	50.0	62.5	56.2	56.2	56.2	56.2	56.2	62.5	68.8
EU-country median	6.2	12.5	9.4	12.5	18.8	18.8	15.6	12.5	18.8	18.8	12.5	12.5	18.8	12.5	12.5	18.8	12.5

Before the crisis, four of them did — about a quarter. Now eleven do — about two thirds. And they are not sixteen ways of saying the same thing: pay per hour, hours worked, what households manage to save, what they expect of the coming year, the real value of the poverty line itself, and the number of citizens packing up and leaving all sit down there together.

This does not explain anything. We tried to use it as an explanation and it failed — on its own it predicts nothing, and put alongside the other measures of accumulated damage it flips sign, which is what a number does when it is describing the weather rather than causing it. What we didn't test returns to that.

What it does settle is smaller and worth having. The measure that put Greece at the top of Europe is not one strange instrument twitching on its own. It is sitting in the middle of a crowd of measures that all moved the same way.

What money actually buys

Income is not the same as what a household can get for it. Two countries can report similar wages and offer very different lives, depending on what things cost.

The measure that captures this counts what people in a country actually consume, adjusted for what things cost there — not what they nominally earn. It is the closest thing in official statistics to asking what a household can actually get.

On that measure Greece has risen substantially over the decade, from roughly 14,800 to 21,300 in the units used for these comparisons. That is a real improvement and it is worth saying clearly, because a report about hardship can give the impression that nothing improved. A great deal improved.

And the measure still predicts reported hardship after income poverty has been accounted for — meaning it carries information the official poverty rate does not.

FIGURE 4 **Material resources rose substantially in Greece, but the gap to the EU median widened**

PRE-PLANNED CONFIRMATORY

How did Greece's material resources move against the EU median?

Material resources

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece	14770	14786	15372	15833	16372	15475	16772	19402	20701	21310
EU median	16230	16494	17038	18006	18707	17386	19479	21317	23055	24129

> Material resources predicts hardship beyond AROP and year effects (coef -0.0013, wild-cluster bootstrap p=0.0055).
>
> *The limits of this.* cross-country association | no causal claim.

Read that carefully: it says the measure carries information the official poverty rate does not. It does not say that raising it would lower hardship by a calculable amount. Nothing in this study can establish that. What it establishes is that a country's real consuming power tells you something about how its households answer the question, over and above where they sit in their own income distribution.

When joblessness stops being a spell

The headline unemployment rate tells you how many people are out of work this month. It tells you nothing about whether those are the same people as last year.

That distinction is not a technicality. A labour market where people lose jobs and find new ones within months is a different place from one where the same people have been out of work for years, even if both report the same percentage. The first is churn. The second is a condition.

Long-term unemployment — out of work for twelve months or more — measures the second thing, and Greece's figure was extraordinary. In 2015 it stood at 16.4% of the labour force. At the worst of it, roughly three out of every four unemployed Greeks had been out of work for over a year. Not between jobs. Out.

By 2024 it had fallen to 5.4%. That is substantial, real progress, and it is still among the highest in Europe — most of the EU sits under 2%.

The reason this matters for a question about making ends meet is that time does something specific to a household. Savings go first, then whatever could be sold, then the goodwill of relatives, then the ability to absorb any surprise at all. A month of unemployment is an inconvenience for most households. Two years is a different category of event, and the household that comes out the other side is not the household that went in.

FIGURE 5 **Long-term unemployment fell sharply and the gap to the EU median narrowed, though Greece remains several times higher**

PRE-PLANNED CONFIRMATORY

How did Greece's long-term unemployment move against the EU median?

Long-term unemployment

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece	16.4	15.4	14.3	12.5	11.3	10.5	9.2	7.7	6.2	5.4
EU median	3.6	3.1	2.7	2.1	1.9	1.8	2.0	1.9	1.7	1.7

> Labour-market exclusion predicts hardship beyond AROP and year effects (coef +4.3402, wild-cluster bootstrap p=0.0085).
>
> *The limits of this.* cross-country association | no causal claim.

This is the clearest of the present-day results, and it survives every check applied to it, including being re-estimated twenty-seven times with a different country left out each time. Dropping Greece itself does not overturn it — which matters, because a finding that only held because of the country the report is about would be worthless.

A paycheck that never came back

Of everything measured here, this is the series that has not recovered.

Greek real wages stood at about 77% of their 2008 level in 2015. In 2024 they stood at about 68%. Not recovering slowly — not recovering. Greek wages have now been below their pre-crisis level for fifteen consecutive years, longer than any country in the EU except Hungary.

What matters for households is not the wage alone but the wage against what things cost, and that combination also predicts reported hardship beyond income poverty.

FIGURE 6 **Wage-adjusted affordability got worse in Greece while the EU median eased**

PRE-PLANNED CONFIRMATORY

How did Greece's wage-adjusted affordability move against the EU median?

Wage-adjusted affordability

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece	141.9	148.6	148.2	156.8	162.2	163.3	165.6	171.2	173.1	173.1
EU median	127.2	124.7	124.6	119.8	118.4	117.7	120.2	120.7	122.3	121.0

> Wage-adjusted affordability predicts hardship beyond AROP and year effects (coef +0.3110, wild-cluster bootstrap p=0.0005).
>
> *The limits of this.* cross-country association | no causal claim.

A country can score badly here two ways: by being expensive, or by paying poorly. Greece does both at once, and the household experiencing it does not much care which half is responsible.

It is worth sitting with what fifteen years means. A worker who started a job in 2008 at the going rate has spent an entire career — the whole of their thirties, say — without the pay level they began with ever returning. Not a bad year. Not a bad stretch. The entire span in which most people expect their earnings to rise.

Set this beside long-term unemployment's own recovery and you have the shape of the Greek recovery as a whole. A household looking at a labour market that has genuinely improved is also looking at a wage that has been below where it started for a decade and a half.

> *“The unemployment rate came back. The paycheck did not.”*

DESCRIPTIVE CORROBORATION

INDEPENDENT DESCRIPTIVE CORROBORATION (GREECE IN FIGURES)

An outside analysis (Greece in Figures, working from newer Eurostat and ELSTAT numbers than the ones used here) lands on the same shape from a different direction and without running any of the tests in this report: Greece is not last in the EU on what people actually consume, but Greek employees put in some of the longest hours in the Union for comparatively low pay per hour, and everyday things like food cost more than that pay would suggest. Same picture, different route, no formal test behind it.

What this lets us say. The independent picture -- Greece not last on actual consumption, long hours for comparatively low hourly reward, high relative prices in food and information/communication -- is consistent with and external corroboration for this project's own material-resources and wage-adjusted-affordability findings. Its central descriptive numbers reproduce against the primary Eurostat and ELSTAT releases they draw on.

What it does not. Treating the article's own comparisons, or its constructed AIC-per-hour ratio, as inferential evidence: it applies no bootstrap, no FDR correction, no within/between decomposition and no model comparison, and it does not test whether these factors account for reported hardship. Merging its 2025/2026-vintage figures into this project's frozen 2015-2024 panel. Repeating its car-ownership sentence, which cites a vehicle-STOCK figure as evidence of new purchases, or its 'all Greeks travelled' generalisation, which the underlying ELSTAT survey does not support. Reading its AIC rank, hours rank or price-level figure as contradicting this project's own numbers without first noting the different vintage or aggregate behind each.

Greece in Figures, "Γιατί οι Έλληνες νιώθουν τόσο φτωχοί" [Why Greeks feel so poor]. source

THE PAST REMAINS PRESENT

How long it went on

Two countries can look identical today and have arrived by different roads. One has had high long-term unemployment for a year. The other has had it for twelve. The intuition that these are not the same situation is strong, and it is the intuition behind almost every account of the Greek crisis.

Intuition isn't evidence, and this is the chapter where an appealing story is easiest to oversell. Three drafts of this report oversold it before the wording below was settled.

For each measure that worked in the earlier chapters, we built a matching one that counts not the level today but the total burden a country has absorbed since before the crisis: how much excess unemployment it accumulated, how many consecutive years its wages stayed below 2008, how much its housing costs deteriorated.

FIGURE 7 **Greece ranks first on accumulated unemployment and housing deterioration, and second on wage non-recovery**

POST-SELECTION ROBUSTNESS

How much accumulated unemployment has each country absorbed, and where does Greece sit?

▼ **Methods and limits**

Read with this. cross-country association | no demonstrated within-Greece dynamic | no causal claim | current conditions not ruled out The three measures are in different units and may not be compared with each other: accumulated unemployment and housing deterioration are percentage-point-years, wage non-recovery is a count of consecutive years.

Accumulated unemployment

Country	Accumulated unemployment (percentage-point-years above 2009)
Greece	137.5
Cyprus	64.1
Croatia	43.5
Spain	33.1
Italy	32.9
Bulgaria	26.6
Portugal	20.6
Slovenia	20.6
Netherlands	13.1
Luxembourg	9.1
Ireland	8.9
Slovakia	8.9
Poland	7.7
France	5.9
Lithuania	5.6
Denmark	5.6
Hungary	3.2
Estonia	3.1
Finland	2.9
Austria	2.5
Belgium	2.4
Romania	2.4
Latvia	2.0
Czechia	1.2
Sweden	0.6
Malta	0.0
Germany	0.0

Years wages below 2008

Country	Years wages below 2008 (consecutive years below the 2008 level)
Hungary	16.0
Greece	15.0
Italy	14.0
Finland	3.0
Sweden	2.0
Slovenia	0.0
Romania	0.0
Portugal	0.0
Poland	0.0
Netherlands	0.0
Malta	0.0
Latvia	0.0
Luxembourg	0.0
Lithuania	0.0
Austria	0.0
Ireland	0.0
Belgium	0.0
Croatia	0.0
France	0.0
Spain	0.0
Estonia	0.0
Denmark	0.0
Germany	0.0
Czechia	0.0
Cyprus	0.0
Bulgaria	0.0
Slovakia	0.0

Housing deterioration since 2010

Country	Housing deterioration since 2010 (percentage-point-years above 2010)
Greece	233.2
Bulgaria	116.3
Luxembourg	37.8
Portugal	35.8
Sweden	17.1
Estonia	11.8
Slovenia	9.9
Hungary	9.6
Germany	9.4
Italy	8.2
Belgium	7.9
Finland	7.0
France	6.6
Slovakia	6.1
Latvia	5.7
Netherlands	4.9
Ireland	4.6
Spain	4.5
Malta	4.5
Austria	4.3
Poland	4.2
Romania	4.2
Czechia	3.8
Cyprus	2.1
Lithuania	0.5
Croatia	0.0
Denmark	0.0

Greece has absorbed a great deal on these measures. On one of the three it is not the highest — Hungary has a longer run of depressed wages — and the chart shows every country rather than Greece alone, so that this is visible rather than buried.

Showing that Greece accumulated a lot is description. The real test is harder: does the history predict hardship *after* you already know the present-day situation? If you know today's unemployment rate, does the past decade of it still add anything?

For three measures, it does.

> Accumulated excess unemployment predicts hardship (bootstrap p=0.0025) and retains a cross-country association after its current-level counterpart is controlled (conditional bootstrap p=0.0025).
>
> *The limits of this.* cross-country association | no demonstrated within-Greece dynamic | no causal claim | current conditions not ruled out.

> Duration of real wages below their 2008 level predicts hardship (bootstrap p=0.0245) and retains a cross-country association after its current-level counterpart is controlled (conditional bootstrap p=0.0060).
>
> *The limits of this.* cross-country association | no demonstrated within-Greece dynamic | no causal claim | current conditions not ruled out | supported only as the CURRENT uninterrupted run against a fixed 2008 base; alternative constructions point the same way but do not meet the criteria.

> Housing-cost deterioration since 2010 predicts hardship (bootstrap p=0.0460) and retains a cross-country association after its current-level counterpart is controlled (conditional bootstrap p=0.0015).
>
> *The limits of this.* cross-country association | no demonstrated within-Greece dynamic | no causal claim | current conditions not ruled out | BORDERLINE: bootstrap p=0.0460, 91 of 1,999 exceedances.

The third of those is borderline and is labelled so rather than rounded up. The second holds only for one specific way of counting, and other reasonable ways of counting the same idea point the same direction without meeting the standard — they are not counted as support.

And the pattern is not universal, which is itself informative.

> For wage-adjusted affordability the pattern reverses: the current measure survives conditioning (bootstrap p=0.0035) while the accumulated one remains inconclusive.
>
> *The limits of this.* the accumulated measure is inconclusive, NOT unsupported.

For the cost-of-living measure it runs the other way: today's number survives and the accumulated one doesn't resolve. If history mattered as a general rule, that reversal shouldn't happen. It suggests these results are specific to work, wages and housing rather than reflecting some broad law that the past always counts.

One measure couldn't be tested at all, and is reported rather than quietly dropped.

> Accumulated material resources (C1) could not be tested at all: the source series begins in 2015 and no 2008 baseline exists. The baseline was not moved to make it testable.
>
> *The limits of this.* infeasible, not null.

The data for it simply starts too late. Moving the starting line to make it testable would have turned it into a measure of something else — one that couldn't see the crisis it was built to see. The starting line stayed where it was.

> The accumulated wage-shortfall coefficient cleared every conditional gate but its prior stage did not support it, so the pre-registered ceiling caps it. It is reported and is not a finding.
>
> *The limits of this.* capped by the E7 ceiling: E7 may only qualify or withdraw.

And one result passed every test in this chapter without being counted, because the measure it was paired with hadn't been supported in the earlier round, and a rule set in advance stops this stage from promoting anything its predecessor didn't establish. On the face of the numbers it looks as convincing as the three that count. The rule held anyway, which is the only condition under which having rules means anything.

WHAT THE EVIDENCE CAN'T SETTLE

Silence isn't a verdict

Nine present-day measures were tested. Three worked. This chapter is about the other six, and it is more important than it sounds.

The tempting thing to write is that the six were ruled out. For most of them, that would be false.

> Six of nine current-level constructs are inconclusive under available power, not unsupported; two cleared FDR and collapsed under the bootstrap (p=0.40 and 0.55).
>
> *The limits of this.* inconclusive is not evidence of absence.

A test that finds nothing tells you something only if it was capable of finding something. With twenty-seven countries and about a decade of data, this study can detect large effects and not small ones. For most of the six, the smallest effect it could reliably have caught is bigger than any effect worth caring about. So the study is *silent* about them. Not negative — silent. Reporting that as "no evidence of an effect" would be one of the easiest and most common ways to mislead with a true sentence.

A few are genuinely different, and for those the exclusion is real — though it holds at a stated size rather than in general.

> Annual food and housing inflation are unsupported with adequate power at 0.70 SD; annual headline inflation is unsupported at its detectable conditional magnitude; compounded inflation since 2008 remains inconclusive.
>
> *The limits of this.* the exclusions are narrow and magnitude-specific | compounded inflation is INCONCLUSIVE, not ruled out.

Inflation is the clearest case. Two inflation measures can be ruled out at the size this study could detect. That is not the same as showing that rising prices don't matter to Greek households, and it should not be read that way.

One more thing belongs here, because it shows why the checks were worth having. Two measures passed the standard statistical test comfortably and then failed a stricter one designed for studies with few countries. Had the protocol stopped at the usual step, this report would contain two more findings than it does.

A photograph, not a film

Everything in the accumulated-history chapter is a statement about how countries differ from each other. It is very tempting, and it is wrong, to turn it into a statement about how Greece changed over time.

> No accumulated measure permits dynamic wording. Across P5, E4 and E7 no within-country estimate is significant in the adverse direction and no first-difference test supports one.
>
> *The limits of this.* three RELATED checks on one panel, not independent replications | the dynamic tests were not multiplicity-corrected.

Here is the distinction, because it is the single easiest thing in this report to get wrong. "Countries that accumulated more hardship report more difficulty" is a claim about a group photograph. "As Greece accumulated more hardship, Greek households reported more difficulty" is a claim about a film. This study took the photograph. It did not shoot the film.

We tested for the film anyway, three separate ways, and found nothing that supports it. And here the caution runs in the other direction: finding nothing is not the same as showing there is nothing. Those tests throw away all the comparison between countries and rely only on movement within each one, which leaves far less to work with. They are too imprecise to settle the question either way.

So: the honest sentence is that this design cannot support a claim about Greece changing over time. Not that Greece didn't change.

This is an unsatisfying place to leave a chapter and it is where the evidence stops. The reason to say it this precisely, rather than letting the group photograph quietly stand in for the film, is that the film is the version everybody wants — it is the version that sounds like an explanation. It is also the version this study did not produce.

Third place, or twenty-fifth?

Most reports include a section showing their findings hold up. This one includes a section showing that one of them doesn't, and that is a result rather than an embarrassment.

Remember from earlier that the deprivation items — can't pay bills, can't heat the home — absorbed most of Greece's unexplained excess. And remember that those items come from the same survey as the thing being explained.

Whether to let a measure like that into the model is a real choice with a real case on both sides. For: it is a meaningful, officially published measure of hardship, and refusing to use variables because they are well measured is not a principle anyone actually holds. Against: when the question and the answer come from the same person in the same interview, some of the agreement between them is the interview rather than the world.

Both positions are defensible. So we ran both.

> Greece's residual is +6.93 (rank 3/27) in the main model and -9.39 (rank 25/27) when the same-instrument deprivation predictor is removed, on identical rows. Neither specification is definitive.
>
> *The limits of this.* NEITHER specification is definitive | the two may not be merged or averaged | selection between them may not be made on residual size | conclusions about absorption depend materially on whether same-instrument deprivation is admitted.

FIGURE 8 **Removing the same-instrument deprivation measure moves Greece from under-predicted to over-predicted**

POST-SELECTION ROBUSTNESS Does the answer depend on which model is chosen?

▼ **Methods and limits**

Read with this. NEITHER specification is definitive. They may not be merged or averaged, and selection may not be made on residual size.

Country	Reference specification	Deprivation-free companion	Change
Luxembourg	8.80	15.48	6.68
Cyprus	7.05	10.75	3.70
Greece	6.93	-9.39	-16.32
Latvia	6.32	4.65	-1.66
Hungary	4.86	7.42	2.56
Bulgaria	4.06	14.65	10.60
Czechia	3.53	-3.63	-7.16
Slovakia	3.28	-0.68	-3.96
Belgium	3.13	5.26	2.13
Ireland	3.05	6.14	3.09
Croatia	2.62	-0.38	-2.99
France	1.85	3.88	2.02
Austria	1.36	0.95	-0.41
Portugal	1.30	0.62	-0.67
Malta	1.28	0.36	-0.91
Poland	0.56	-4.32	-4.88
Denmark	0.16	-3.65	-3.81
Sweden	-1.00	-4.55	-3.55
Slovenia	-2.66	-6.33	-3.67
Netherlands	-3.47	-4.87	-1.39
Lithuania	-4.17	-1.81	2.36
Germany	-4.31	-6.00	-1.69
Estonia	-4.49	-10.12	-5.63
Finland	-5.07	-6.80	-1.72
Italy	-6.35	-5.18	1.17
Romania	-10.32	9.76	20.08
Spain	-10.63	-10.06	0.57

Greece moves from the third-worst country in Europe on unexplained hardship to the twenty-fifth — from a stark positive outlier to a stark negative one — on exactly the same rows of data, with one variable added or removed. Third to twenty-fifth, out of twenty-seven, from a single judgement call.

> *"That is not a wobble. It is a reversal, and there is no honest way to pick between the two."*

We can't average them. We can't choose the one that looks more plausible, because choosing the model that gives the answer you expected is how this kind of check gets rendered meaningless. What can be said is this: how much of Greece's anomaly gets absorbed depends on a judgement call the data cannot settle, and any conclusion resting on that absorption inherits the uncertainty.

Which is why the conclusion of this report does not rest on it.

The chart we didn't publish

Two analyses were planned in advance, meant to carry real weight, and neither worked. They are here because a report that shows only its successes gives a false impression of how much was tried.

The first was the one that would have made the best chart in this report. The idea is elegant: build an artificial Greece out of a weighted blend of other countries, one that tracks the real Greece closely before 2008, then watch the two diverge afterwards. The gap between them is the crisis.

It failed. The blend collapsed onto essentially two countries — Hungary and Bulgaria — which is not a stand-in for Greece, and the design missed four of the six conditions set for it in advance.

> The synthetic-control comparative design failed four of six pre-registered gates and is not a usable comparison. Its divergence figure is kept out of every document.
>
> *The limits of this.* donor weights collapse to Hungary 0.55 and Bulgaria 0.45 | the divergence figure is NON-REPORTABLE.

The chart it would have produced is not in this report, and that is deliberate. A dramatic picture resting on a counterfactual that thin would convince people of something the evidence cannot support. Being convincing and being right are different properties, and the first is most dangerous when the second is missing.

The second was a measure of how many different kinds of disadvantage a country shows at once. Adding it made Greece's position worse rather than better, and its direction flipped once other things were held constant. That flip is left unexplained here on purpose: inventing a story for a strange result in an analysis that has already failed is exactly how failed analyses come back from the dead.

> Multi-domain breadth failed the incremental criterion: adding it to the main model worsened Greece's residual from 6.93 to 10.39 and reversed its sign conditionally. The reversal is left uninterpreted.
>
> *The limits of this.* a result about the specification, not about power | the sign reversal is deliberately left uninterpreted.

Just gloomier answerers?

There is a simpler explanation for everything so far, and it deserves to be taken seriously rather than waved away: maybe Greeks are just gloomier answerers.

Earlier got partway to an answer — the reported difficulty does track real material trouble — but all of that evidence came from inside one survey, so it can't settle this on its own.

A better test looks across *different* subjects. A general tendency to answer darkly should drag everything down about equally. A pattern that is extreme on money and milder elsewhere points at circumstances instead.

MEASURE	GREECE	EU MEDIAN	GREECE'S POSITION
Reported hardship	66.7%	17.1%	1st of 27, worst first
Financial expectations	-43.2	-1.8	1st of 27, worst first
Life satisfaction	6.7	7.3	2nd of 28, worst first

> Greece is consistently among Europe's worst countries on life satisfaction and consistently worst on financial hardship and expectations. The greater extremity of the financial indicators suggests domain specificity, but does not rule out a broader negative reporting tendency.
>
> *The limits of this.* Greece is second-WORST on life satisfaction by 2024, not middling | a broader negative reporting tendency is NOT ruled out | Greek life satisfaction ROSE over the observed period, 6.2 to 6.9; only its rank worsened, because other countries improved faster | the series begins in 2013, at the crisis trough, with no pre-crisis baseline.

Each row compares Greece with the EU-country median directly, and gives Greece's rank against the twenty-seven or so other member states that report it. The pattern is specific to money, but it is a difference of degree, not of kind. Greece is worst in Europe on the financial questions and close to worst on general life satisfaction. That is not the profile of a country that is desperate about money and otherwise content, and an earlier version of this report described it that way and was wrong.

One more thing has to be held apart here, because getting it backwards inverts the finding. Greek life satisfaction *rose* over this period, from 6.2 to 6.9. Greece's *rank* got worse anyway, because other countries improved faster. A falling rank on a rising number is not a country getting unhappier.

DESCRIPTIVE CORROBORATION

FINANCIAL EXPECTATIONS AND LIFE SATISFACTION

Greece's money questions sit at the far edge of the European range while its general-wellbeing question sits closer to it. That difference of degree is what the comparison shows, and it is description rather than a test.

What this lets us say. The financial indicators are more extreme than the general-wellbeing one, which suggests domain specificity. A broader negative reporting tendency is NOT ruled out.

What it does not. Describing Greece as ordinary or middling on life satisfaction: it is second-worst in the EU by 2024. And reading a worsening RANK as falling satisfaction, when the Greek level rose over the period.

reporting_style_cross_indicator.csv, this project

Already behind, before 2008

There is a hole in the middle of the last chapter. The European survey used throughout this report only starts asking about life satisfaction in 2013 — at the bottom of the Greek crisis. So it cannot tell you whether Greece was already unusual *before* 2008, which is exactly what you would want to know.

A different survey can, and what follows is kept deliberately separate: a different instrument, shown on its own, never joined to the other series.

Greek life satisfaction across six ESS rounds, against the median of the same twelve countries in every round. Means are approximate reconstructions from published weighted percentages: no interval can be attached to them and nothing is tested on them.

PERIOD	FIELDWORK	GREECE	MEDIAN OF THE 12	GAP	RANK OF 12
pre-crisis	2002/03	6.35	7.21	-0.86	3
pre-crisis	2004/05	6.46	7.25	-0.80	4
crisis onset	2008/09	5.96	7.09	-1.13	3
early crisis	2010/11	5.64	7.00	-1.36	1 (worst)
later period	2020-22	6.35	7.48	-1.12	1 (worst)
later period	2023/24	6.42	7.39	-0.97	1 (worst)

No Greek round falls between 2010/11 and 2020-22, so the decade covering the depth of the adjustment and the recovery is unobserved.

Three things happen in order. Greece was *already* about 0.8 points below the middle of this group before the crisis. Its level then fell sharply, to 5.64 by 2010/11. And by 2023/24 it had climbed back to 6.42, roughly where it started.

Its position, though, did not climb back. The gap to the middle of the group is wider now than before the crisis, and Greece has been the lowest of these twelve countries since 2010/11 — the two countries that used to sit below it have both passed it.

This cuts against a tidy conclusion, which is why it is here. A low Greek starting point pre-dates the crisis, so the crisis cannot be the whole story, and a long-standing pattern of lower reported wellbeing remains entirely plausible. Gloominess as a national trait can't be dismissed on this evidence. It also can't be established by it.

DESCRIPTIVE CORROBORATION

PRE-CRISIS WELLBEING BASELINE (ESS)

Across six rounds of a separate European survey, holding the same twelve countries fixed each time, Greece's level fell and then recovered while its position relative to the others did not.

This is descriptive corroboration and not a test: it shows a pattern that sits alongside the previous chapter without confirming or refuting it.

What this lets us say. On a balanced set of 12 countries observed in every Greek ESS round, Greece already sat about 0.8 points below the median before the crisis, fell to 5.64 by 2010/11, and recovered its LEVEL to roughly the pre-crisis value by 2023/24. Its comparative position did not recover: the gap to the balanced median is wider than before the crisis, and Greece has been worst of the 12 since 2010/11. A longstanding low-wellbeing pattern therefore remains plausible and generic pessimism or reporting culture CANNOT be dismissed.

What it does not. Splicing ESS to the Eurostat EU-SILC series, or reading the two as one trajectory. Attaching any confidence interval, standard error or significance test to these means, which are approximate reconstructions from displayed weighted percentages. Comparing ALL-COUNTRY ranks across rounds, since the country set varies from 22 to 30. Treating the decade after 2010/11 as observed.

European Social Survey Data Portal, public Analysis tab for stflife (life satisfaction, 0-10), rounds 1, 2, 4, 5, 10 and 11. source

A caution about where these numbers come from: they are reconstructed from percentages the survey publishes publicly, not from the underlying responses, which sit behind a login. They are approximate, they carry no margin of error, and nothing is tested on them. There is also a decade with no Greek survey round at all, right through the deepest part of the adjustment.

What this report leaves out

Several things that come up in every conversation about the Greek crisis are not established in this report. Leaving them out silently would be misleading. Discussing them as though they were findings would be worse.

Some of them were looked at. Migration was tested here directly. The cross-country comparison and the earlier survey a little further back are both analyses done for this report. What none of them can do is carry a conclusion, and each one below says what it permits and what it doesn't.

CONTEXTUAL EVIDENCE

INSTITUTIONAL TRUST

Trust in institutions is low in Greece, and there is a plausible route by which it could matter: a household that doesn't expect help to arrive may experience the same circumstances as more frightening. Nothing here tested that.

What this lets us say. May affect how households interpret insecurity; available evidence does not identify an independent effect.

What it does not. Presenting trust as an explanation of the residual, or implying it was tested and found to matter.

OECD (2024), OECD Survey on Drivers of Trust in Public Institutions - 2024 Results, Country Notes: Greece. OECD Publishing, Paris. source

LITERATURE-GROUNDED CONTEXT

CRISIS AND ADJUSTMENT POLICIES

The bailout programmes from 2010 reshaped incomes, job protections, pensions and public services at once and in a hurry. They are the backdrop to every accumulated measure described earlier.

What this lets us say. Help explain the historical setting; this project does not estimate their causal contribution.

What it does not. Attributing any share of the accumulated exposure to a specific programme or measure.

Andriopoulou, E., Kanavitsa, E. & Tsakloglou, P. (2020), Decomposing Poverty in Hard Times: Greece 2007-2016. LSE GreeSE Paper No. 149. source

CONTEXTUAL CONSEQUENCE

MIGRATION

Large numbers of working-age Greeks left during the crisis, and some have returned. This cuts both ways: plausibly a consequence of a broken labour market, and plausibly part of why the people who stayed look the way they do.

What this lets us say. A plausible consequence of prolonged labour-market damage and a possible contributor to it. It is UNSUPPORTED as an independent aggregate predictor here.

What it does not. Reading it as a driver, or as ruled out. E3 tested it on this panel and found nothing (p=0.4006), which speaks to aggregate prediction and not to either causal direction.

Lazaretou, S. (2016), The Greek brain drain: the new pattern of Greek emigration during the recent crisis. Economic Bulletin, Bank of Greece, issue 43, pp. 31-53. source

FUTURE HYPOTHESIS

TAX BURDEN AND UNEQUAL TREATMENT

Published research establishes that Greece's system of indirect taxes became markedly harder on lower-income households across the crisis. If a household's position worsened through tax in a way income-based poverty measures capture badly, that would be one route to the kind of gap this report describes.

What this lets us say. Published incidence evidence establishes that Greece's indirect tax system became markedly more regressive over the crisis. Whether that channel contributes to the hardship gap is a plausible hypothesis and is NOT tested here.

What it does not. Any quantitative statement linking tax burden to the hardship gap. The literature is about incidence, not about this outcome, and this project tested nothing on tax.

Kaplanoglou, G. (2015), Who Pays Indirect Taxes in Greece? From EU Entry to the Fiscal Crisis. Public Finance Review 43(4), 529-556. source

What this adds up to

Greek households report struggling at a rate far above what the official poverty figure predicts, and they have done so consistently for a decade. Three bodies of evidence make part of that distance easier to understand.

The official measures are narrower than the experience they get used to summarise. The wider net closes about a fifth of the gap and is closing less each year, and the income line itself fell along with the Greek economy, which a relative measure cannot help doing.

What households report corresponds to real material trouble — bills, heating, unexpected expenses — though that agreement comes from inside a single survey rather than from an independent source.

And both the present and the past carry information: what a country can actually afford, how much long-term unemployment it carries, how its wages stand against its prices, together with the accumulated weight of unemployment, of wages that never recovered, and of housing costs that worsened.

What the evidence does not support is just as much part of the answer. Nothing here shows what causes what. Nothing here shows Greece changing over time; that question was asked and the answer was too imprecise to settle. Most of the measures that didn't work are unresolved rather than ruled out. One central result flips depending on a judgement call the data can't settle. And a general tendency to answer darkly cannot be excluded — the survey reaching back before the crisis suggests Greece started lower.

And most of the gap is still unexplained. Of the 52.6 points, what is accounted for here is a minority. The rest is not attributed to anything, because we could not establish where it goes.

AUTHOR INTERPRETATION

POLICY IMPLICATIONS

What runs through all of this is that no single official number captures what Greek households are reporting. Each one misses something different, and the ones that catch what the others miss are not the headline.

What this lets us say. POLICY RECOMMENDATION, NOT AN EMPIRICAL CONCLUSION: poverty dashboards should combine AROP, its real threshold, anchored poverty, AROPE/deprivation and accumulated labour and housing indicators.

What it does not. Presenting this as a finding. It follows from what the analysis showed AROP alone misses; it is not itself a result.

That is a recommendation about how poverty gets reported, and it follows from what the single measures were seen to miss. It is not itself a finding, and it is the last thing in this report rather than the first for that reason.